

COURSE TITLE : DIGITAL COMPUTING LAB
COURSE CODE : 4268
COURSE CATEGORY : A
PERIODS/WEEK : 6
PERIODS/SEMESTER : 90
CREDITS : 3

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Basic Logic Gates	22
2	Combinational circuit design	23
3	Sequential Circuit design	22
4	8086 Microprocessor programming	23

COURSE GENERAL OUTCOMES:

Sl.	G.O	On completion of this course the student will be able to :
1	1	To understand basic logic gates
2	1	To design combinational logic circuits
3	1	To design sequential logic circuits
4	1	To practice programming with 8086 microprocessor

SPECIFIC OUTCOMES:

CYCLE – I: Basic Logic Gates

1.1 To understand basic logic gates

1.1.1 Demonstrate basic logic gate ICs.

1.1.2 Apply Karnaugh map to minimize logic functions

1.1.3 Construct basic gates(NOT,OR,AND,XOR,XNOR) using NAND gates

1.1.4 Construct basic gates(NOT,OR,AND,XOR,XNOR) using NOR gates

CYCLE – II: Combinational circuit design

2.1 To design combinational logic circuits

- 2.1.1** Demonstrate basics of combinational logic circuit design
- 2.1.2** Design and Implement Adders (Half adder, Full adder, Ripple carry adder, Carry Look Ahead adder)
- 2.1.3** Design and Implement Subtractors (Half Subtractors, Full subtractors)
- 2.1.4** Design and Implement Multiplexers, Demultiplexers.
- 2.1.5** Design and Implement Encoders, Decoders.

CYCLE – III: Sequential Circuit design

3.1 To design sequential logic circuits

- 3.1.1** Demonstrate basics of sequential circuit design
- 3.1.2** Implementation of basic flip flops (RS, JK, D, T, Master-Slave)
- 3.1.3** Design and implementation of Synchronous (Ring counter, Johnson counter) and Asynchronous Counters (UP/DOWN counters).
- 3.1.4** Design and implementation of Shift Registers

CYCLE – IV: 8086 Microprocessor programming

4.1 To practice programming with 8086 microprocessor

- 4.1.1** Demonstrate 8086 microprocessor training kit
- 4.1.2** Create simple programs (without jump statements) with 8086 training kit.
- 4.1.3** Create programs having conditional jump with 8086 training kit.